

BIF501

BIOINFORMATICS-II

ASSIGNMENT NO. 1
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- 01 Describe Five DNA Sequence Formats By Giving Examples
- 02 Write Methods For Retrieving DNA Data From NCBI
- 03 Describe Three DNA Databases

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Assignment Questions

1. Describe Five DNA Sequence Formats By Giving Examples
2. Write Methods For Retrieving DNA Data From NCBI
3. Describe Three DNA Databases

Solution

1. Describe Five DNA Sequence Formats By Giving Examples

I. FASTA Format

In bioinformatics, FASTA format is a text-based format for representing DNA sequences, in which base pairs are represented using a single-letter code [A,C,G,T,N] where A=Adenosine, C=Cytosine, G=Guanine, T=Thymidine and N= any of A,C,G,T. The format also allows for sequence names and comments to precede the sequences.

A sequence in FASTA format begins with a single-line identifier description, followed by lines of DNA sequence data. The identifier description line is distinguished from the sequence data by a greater-than (>) symbol in the first column. The word following the ">" symbol is the identifier of the sequence, and the rest of the line is a description (optional) separated from the identifier by a white space or tab. The sequence data starts on the next line following the text line and ends if another line starting with a ">" appears; this indicates the start of another sequence.

Examples of the FASTA format:

```
>P01013 GENE X PROTEIN (OVALBUMIN-RELATED)
QIKDLLVSSSTDLDTTLVLVNAIYFKGMWKTAFAEDTREMPPHVTQESKPVQMMCMNNSFNVAT
LP AE
KMKILELPPFASGDL SMLVLLPDEVSDLERIEKTINFEKLT EWTNPNTMEKRRVKVYLPQM KIEEKY
NLTS
VLMALGMTDLFIP SANLTGISSAESLKISQAVHGAFMELSEDGIEMAGSTGVIEDIKHSPES EQFR
ADHP
FLFLIKHNPTNTIVYFGRYWSP
```

II. GeneBank Sequence Format

GeneBank sequence format is found to be in the GeneBank Database which is a kind of standard format and other formats are pretty similar to it. GenBank format (GenBank Flat File Format) stores sequence and its annotation together. The start of the annotation section is marked by a line beginning with the word “LOCUS”. The start of sequence section is marked by a line beginning with the word “ORIGIN” and the end of the section is marked by a line with only “//”.

Example of the GeneBank Sequence Format:

```
LOCUS      AB000263                368 bp    mRNA    linear    PRI 05-FEB-1999
DEFINITION Homo sapiens mRNA for prepro cortistatin like peptide, complete
            cds.
ACCESSION  AB000263
ORIGIN
      1 acaagatgcc attgtccccc ggctcctgc tgctgctgct ctccggggcc acggccaccg
     61 ctgccctgcc cctggagggt ggccccaccg gccgagacag cgagcatatg caggaagcgg
    121 caggaataag gaaaagcagc ctctgactt tcctcgcttg gtggtttgag tggacctccc
    181 aggccagtgc cgggcccctc ataggagagg aagctcgga ggtggccagg cggcaggaag
    241 gcgcaccccc ccagcaatcc gcgcgccggg acagaatgcc ctgcaggaac ttcttctgga
    301 agaccttctc ctctgcaaaa taaaacctca cccatgaatg ctcacgcaag ttaattaca
    361 gacctgaa
//
```

III. EMBL Format

EMBL format stores sequence and its annotation together. The start of the annotation section is marked by a line beginning with the word “ID”. The start of sequence section is marked by a line beginning with the word “SQ”. The “//” (terminator) line also contains no data or comments and designates the end of an entry.

Example of the EMBL Format :

```
ID  AB000263 standard; RNA; PRI; 368 BP.
XX
AC  AB000263;
XX
DE  Homo sapiens mRNA for prepro cortistatin like peptide, complete cds.
XX
SQ  Sequence 368 BP;
```

```

acaagatgcc attgtccccc ggccctcctgc tgctgctgct ctccgggggcc acggccaccg      60
ctgccctgcc cctggagggt ggccccaccg gccgagacag cgagcatatg caggaagcgg      120
caggaataag gaaaagcagc ctctgactt tctcgcttg gtggtttgag tggacctccc      180
aggccagtgc cggggcccctc ataggagagg aagctcggga ggtggccagg cggcaggaag      240
gcgcaccccc ccagcaatcc gcgcgccggg acagaatgcc ctgcaggaac ttcttctgga      300
agaccttctc ctctgcaaa taaaacctca cccatgaatg ctcacgcaag ttaattaca      360
gacctgaa                                     368

```

//

IV. GCG format

A sequence file in GCG format contains exactly one sequence, begins with annotation lines and the start of the sequence is marked by a line ending with two dot (".") characters. This line also contains the sequence identifier, the sequence length and a checksum. This format should only be used if the file was created with the GCG package.

Example of the Sequence In GCG Format Is:

```

ID   AB000263 standard; RNA; PRI; 368 BP.
XX
AC   AB000263;
XX
DE   Homo sapiens mRNA for prepro cortistatin like peptide, complete cds.
XX
SQ   Sequence 368 BP;
AB000263 Length: 368 Check: 4514 ..
      1 acaagatgcc attgtccccc ggccctcctgc tgctgctgct ctccgggggcc acggccaccg
     61 ctgccctgcc cctggagggt ggccccaccg gccgagacag cgagcatatg caggaagcgg
    121 caggaataag gaaaagcagc ctctgactt tctcgcttg gtggtttgag tggacctccc
    181 aggccagtgc cggggcccctc ataggagagg aagctcggga ggtggccagg cggcaggaag
    241 gcgcaccccc ccagcaatcc gcgcgccggg acagaatgcc ctgcaggaac ttcttctgga
    301 agaccttctc ctctgcaaa taaaacctca cccatgaatg ctcacgcaag ttaattaca
    361 gacctgaa

```

V. IG format

A sequence file in IG format can contain several sequences, each consisting of a number of comment lines that must begin with a semicolon (";"), a line with the sequence name (it may not contain

spaces!) and the sequence itself terminated with the termination character '1' for linear or '2' for circular sequences.

Example Sequence In IG Format Is:

```
; comment
; comment
AB000263
ACAAGATGCCATTGTCCCCGGCCTCCTGCTGCTGCTGCTCTCCGGGGCCACGGCCACCGCTGCCCTGCC
CCTGGAGGGTGGCCCCACCGGCCGAGACAGCGAGCATATGCAGGAAGCGGCAGGAATAAGGAAAAGCAGC
CTCCTGACTTTTCCTCGCTTGGTGGTTTGAGTGGACCTCCCAGGCCAGTGCCGGGCCCCCTCATAGGAGAGG
AAGCTCGGGAGGTGGCCAGGCGGCAGGAAGGCGCACCCCCCAGCAATCCGCGCGCCGGGACAGAATGCC
CTGCAGGAACCTTCTTCTGGAAGACCTTCTCCTCCTGCAAATAAAACCTCACCCATGAATGCTCACGCAAG
TTTAATTACAGACCTGAA1
```

2. Write Methods For Retrieving DNA Data From NCBI

- ❖ Search the Gene database with the gene name, symbol. If you know the gene symbol and species, enter them as follows: tpo[sym] AND human[orgn]
- ❖ Click on the desired gene.
- ❖ Click on Reference Sequences in the Table of Contents at the upper right of the gene record.
- ❖ The NCBI Reference Sequences section of the record has links to NCBI curated records for transcripts (NM and XM prefix reference sequences) for the gene of interest for eukaryotic organisms. Transcript sequences are not produced for prokaryotes.
- ❖ If there is no gene record for the organism and gene of interest, perform a search in the UniGene database with the gene name, product name, or symbol.
- ❖ Click on the UniGene cluster of interest.
- ❖ The UniGene cluster has links to transcript sequences for the gene from the Nucleotide and EST databases
- ❖ If there is no UniGene cluster for this gene and organism, perform a search in the Nucleotide database with the gene name, product name, or symbol. Include the organism in the search to find the most relevant results and filter for transcript sequences, for example: Cytochrome c AND bullfrog[orgn] AND mrna[filter].

3. Describe Three DNA Databases

A DNA database or DNA databank is a database of DNA profiles which can be used in the analysis of genetic diseases, genetic fingerprinting for criminology, or genetic genealogy. DNA databases may be public or private, the largest ones being national DNA databases.

1) GenBank

The **GenBank** sequence database is an open access, annotated collection of all publicly available nucleotide sequences and their protein translations. This database is produced at the National Center for Biotechnology Information (NCBI) as part of the International Nucleotide Sequence Database Collaboration (INSDC). GenBank and its collaborators receive sequences produced in laboratories throughout the world from more than 100,000 distinct organisms. GenBank continues to grow at an exponential rate, doubling every 18 months. GenBank is built by direct submissions from individual laboratories, as well as from bulk submissions from large-scale sequencing centers, and submissions from private individuals.

2) EMBL

The EMBL Nucleotide Sequence Database constitutes Europe's primary nucleotide sequence resource. DNA and RNA sequences are directly submitted from researchers and genome sequencing groups and collected from the scientific literature and patent applications. In collaboration with DDBJ and GenBank the database is produced, maintained and distributed at the European Bioinformatics Institute. Database releases are produced quarterly and are distributed on CD-ROM. EBI's network services allow access to the most up-to-date data collection via Internet and World Wide Web interface, providing database searching and sequence similarity facilities plus access to a large number of additional databases.

3) DNA Data Bank of Japan:

The DNA Data Bank of Japan (DDBJ) is a biological database that collects DNA sequences. It is located at the National Institute of Genetics (NIG) in the Shizuoka prefecture of Japan. It is also a member of the International Nucleotide Sequence Database Collaboration or INSDC. It exchanges its data with European Molecular Biology Laboratory at the European Bioinformatics Institute and with GenBank at the National Center for Biotechnology Information on a daily basis. Thus these three databanks contain the same data at any given time.

----- **THANKS** -----